MULTIMODAL AI

PROJECT REPORT

"Offline Multimodal Meeting Intelligence for Decision and Participant Influence Analysis"

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Submitted By

Group 12

Anushka Chatterjee- 23070126015
Amitkumar Racha- 24070126501
Sahil Awatramani- 23070126112
Shriyans Sahoo- 23070126125

UNDER THE GUIDANCE OF

Dr. Nivedita Mishra

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Pune – 412115, Maharashtra State, India

https://www.sitpune.edu.in/

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

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1. Resource Allocation and Risk Analysis

1.1 Resource Allocation

The project will need local storage space, computing power, and various software libraries for processing the multimodal data of meetings.

- **Storage**

  The AMI Meeting Corpus contains a huge amount of audio recordings of the meetings. The raw audio recordings, consisting of headset and microphone array channels, take up more than 80 GB of storage space. There is also the need for additional storage space to store processed audio, transcripts, speaker diarization outputs, extracted presentations, and semantic embeddings.

- **Computational Resources**

  Computing power is needed primarily for the speech-processing parts of the project. Faster whisper is used for Automatic Speech Recognition (ASR). Speaker diarization is performed using the library pyannote.audio [1], [2]. Silero VAD is used for voice activity detection before the speech processing stages [3].

  As ASR and speaker diarization require much computing power, the pipeline of the project is designed to work as an offline modular workflow where the intermediate outputs can be cached and reused instead of performing the same processing on the raw recordings again. ASR also uses int8 quantization for reducing the computational load.

- **Software Resources**

  Python programming language is used as the development environment. Some of the main software libraries used in the pipeline are listed below:

  - Faster-whisper for Automatic Speech Recognition [1]
  - Pyannote.audio for speaker diarization [2]
  - Silero VAD for voice activity detection [3]
  - Python-pptx for extracting presentation content [4]
  - Sentence-transformers for generating semantic embeddings [5]
  - LibreOffice for conversion from legacy .ppt to .pptx format

  These modules are separated into stages to test the modules and reuse their output.

1.2 Risk Analysis and Mitigation

- **Data Format Incompatibility**

Presentation files from the AMI corpus consist of legacy .ppt files; however, the presentation-extraction pipeline requires the .pptx format.

Mitigation: A preliminary step is introduced, where legacy presentation files are batch-converted to .pptx format via headless LibreOffice, followed by extraction of the content.

- **Computational Bottlenecks**

Transcription and diarization of audio are computationally demanding tasks, especially for long recording meetings.

Mitigation: An independent processing pipeline with modular structure and intermediate JSON outputs, like transcripts and alignments, can be cached. This prevents rerunning computationally expensive processing steps during development and testing.

- **Acoustic Challenges**

Overlapping speech, noise, and room reverberation in the recording from a meeting room can influence the performance of Automatic Speech Recognition and speaker diarization.

Mitigation: Speech activity detection and ASR segmentation are aligned by temporal overlap with speaker diarization boundaries. It allows to match a speech segment with its appropriate speaker even if the boundaries from these two tasks do not perfectly match.

- **Multimodal Alignment**

The project consists of combining spoken dialogue and presentation slides. Misalignment in time or semantics between these two streams can impact decision making and influence analysis.

Mitigation: Timestamps for the audio track are preserved through the whole speech processing pipeline, while the content of presentations is translated into textual representations and semantic embeddings. These two types of information can be later aligned based on both temporal and semantic information.

2. Description of the Selected Dataset

This project uses the AMI (Augmented Multi-party Interaction) Meeting Corpus, which is a multimodal database dedicated to the study of meetings and human interaction. The corpus comprises about 100 hours of synchronized meeting recordings made in several instrumented meeting rooms [6].

Most of the meetings consist of the group of four people, playing different roles in the fictional design-project scenario. In addition, the database includes natural meetings, which gives a variety of types of interaction represented in the database [6], [7].

The following multimodal inputs are included in the dataset:

- Close-talking headset microphone recordings of individual participants
- Microphone array recordings of the whole room
- Video streams
- Electronic copies of presentations and documents shown in meetings
- Word-level speech transcripts
- Dialogue-act annotations
- Named-entity annotations
- Meeting summaries
- Decision-related annotations [6], [7]

Synchronization of the sources mentioned above is especially critical for the proposed project since information in one modality can be correlated with information in another one using the timeline of the meeting.

As regards the speech-processing component, word-level transcripts and dialogue-act annotations can be used as reference information for analysing the conversation behavior. Decision-related annotations are appropriate for the evaluation of the decision-analysis component of the proposed system. Presentation and document data contain additional context, which can be correlated with the topics mentioned by participants.

This database has been chosen due to the combination of speech, speaker information, presentation material and manual annotations available in it. It includes all required multimodal inputs and reference data for the evaluation of the proposed offline meeting-intelligence system [6], [7].

3. Modalities Chosen for the Project

The meeting intelligence system proposed is centered around two major modalities: spoken audio and visual/document information. Such modalities were chosen because they complement each other in terms of information available about the meeting: audio captures the interactions between meeting participants, while presentation captures the contextual information about the topic discussed.

3.1 Spoken Audio Modality

Spoken audio modality captures the process of verbal communication between meeting participants. The corpus provided by the AMI dataset includes close-talking headsets and room microphones array audio recordings [6], [7].

Processing of such audio can be decomposed into several steps:

1. Performing Voice Activity Detection (VAD) in order to detect speech regions;
2. Speaker Diarization that will tell us who was speaking at any particular moment in time;
3. Automatic Speech Recognition (ASR) in order to transcribe audio to text;
4. Speaker-attributed transcript generation to link recognized speech to particular participants;
5. Temporal alignment of ASR segments and speaker-diarization segments.

The result will contain the information about what was said, who said it, and when. Such information is essential in analysis of discussion flow, proposals, responses, decision traces, and participant influence.

The speech processing components used are Silero VAD, faster-whisper, and pyannote.audio [1]–[3].

3.2 Visual/Document Modality

The second chosen modality is the presentation and document content provided in the meetings.

The AMI corpus has the digital copy of presentations and documents used in the meetings [6], [7].

This modality is able to provide some information that might not be available from the transcript itself. The presentation content may include:

- Slide text
- Tables
- Notes

- Design alternatives
- Technical details
- Any other material related to the discussion

The first step in the process of document analysis is converting legacy presentation files into format which is suitable for programmatically analysing. Extracting relevant information from these files is done with the help of python-pptx [4]. After extraction this information may be represented by semantic embeddings created with sentence-transformers [5].

This kind of representation allows aligning the presentation content with the corresponding discussion.

3.3 Rationale for Modality Selection

The two modalities provide complementary information:

Modality	Primary Information
Spoken Audio	Who said what, when, and how the discussion progressed
Presentation/Document	What topic, design, or material was being discussed

Combination of these sources will allow the system to analyse the meeting in a way that goes beyond just speech transcription. More specifically this combination will allow performing the tasks of decision analysis and participant influence analysis.

Though the AMI corpus includes video recordings, it was decided not to use this modality for the current implementation of the project.

4. Exploratory Data Analysis (EDA) and Preliminary Data Exploration

The chosen AMI corpus dataset comprises multiple modalities: audio data, presentations, and annotations. As such, EDA was adjusted according to the specifics of each chosen modality. The initial investigation was aimed at exploring the dataset structure, audio properties, conversational annotations, decision annotations, and presentation formats prior to applying the major machine learning components.

4.1 Audio Data Exploration

The audio data was explored both on the file and metadata level to analyze the recording sources available and whether they are appropriate for speech processing. Specifically, the AMI corpus dataset includes both headset audio data and room microphone-array audio data [6], [7].

These two recording settings have distinct properties. On one hand, headset data is more related to individual participants, while room-array data captures the meeting atmosphere and can include background noise, reverberation, and overlapping speech.

In addition, the available WAV audio data recordings were analyzed in terms of their recording length and audio properties. The meeting duration analysis at the corpus level was made using the available audio recordings to assess the variability in meeting lengths. This aspect is important because the system cannot assume all meetings to have identical duration.

The initial analysis showed that audio pre-processing is required prior to ASR and speaker diarization processes. As such, the pre-processing pipeline was implemented to normalize the selected audio input data by converting multi-channel audio to mono and using 16 kHz sampling rate.

4.2 Dialogue-Act Exploration

The AMI data set also features manually annotated dialogue acts, which characterize the various forms of communicative behavior [6], [7]. The annotations were analyzed in order to reveal the linguistic structure of the meetings.

In total, 116,357 annotated dialogue act occurrences were found in the analysis of the dialogue-act XML files. The most common categories and their distributions are shown below:

Dialogue Act	**Count**	**Percentage**
Inform	33,484	28.8%
Backchannel	21,910	18.8%
Assess	21,391	18.4%
Fragment	16,796	14.4%
Suggest	9,391	8.1%
Stall	8,459	7.3%
Elicit-Inform	4,240	3.6%
Other	686	0.6%
Total	**116,357**	**100.0%**

The analysis involved the parsing of the dialogue act XML annotations and conversion of the AMI-specific dialogue-act IDs into dialogue-act categories.

The distribution indicates that the conversation during the meetings contains several types of interactions besides simple informative utterances. In particular, the Inform, Backchannel, Assess, and Fragment categories are prevalent in the annotated conversation, while Suggest and Elicit-Inform categories are related to the proposal and request for information respectively.

It is important for the current project since neither decision formation nor influence of participants on the process can be accurately inferred based on isolated keywords. The role of an utterance within the conversation and its place should be considered as well.

4.3 Decision Annotation Exploration

Decision annotations have also been discovered in the AMI corpus. It is due to the need to look at the structure of decision XML files and understand how decisions are represented in the corpus.

These decision annotations are helpful as reference data for the decision-analysis component that is proposed. The data will help to evaluate the effectiveness of the system in finding important decision points and linking them to the discussion that surrounds the decision.

Analysis suggests that decisions have to be treated in the context of preceding conversational actions. Actions like Suggest and Assess, which are found through the dialogue-act analysis, can provide context clues to follow the progression of the discussion towards the decision.

4.4 Presentation and Document Exploration

Presentation data were explored separately from audio data since they had different formats. The AMI corpus contains digital copies of presentations and documents used in meetings [6], [7].

During the exploration, it was found that there is a problem with format compatibility: presentation files may be in the legacy .ppt format, while the .pptx format is required by the extraction workflow chosen.

In order to solve this problem, document preprocessing workflow has the following steps:

1. Conversion of legacy .ppt files to .pptx using LibreOffice.
2. Textual and structural content extraction of each slide in .pptx using python-pptx [4].
3. Textual preprocessing of the extracted content.
4. Generation of semantic embeddings using sentence-transformers [5].
5. Alignment with the spoken discussion through these representations.

Thus, it was found that document-format conversion is necessary preprocessing stage and not an optional transformation.

4.5 Cross-Modal Data Exploration

Given the fact that the project involves both audio and documents, their relationship was explored during preliminary exploration as well.

The representations of interest can be described as follows:

Audio -> Speech Segments -> Speaker Information -> Transcript

Presentation -> Slide Content -> Semantic Representation

These representations can then be linked by means of temporal and semantic information. For instance, the spoken discussion of some design option can be linked to the presentation content related to the same subject matter.

This intermodal relationship is critical for the decision-lineage part of the project, since the context surrounding the particular decision can be distributed among the spoken conversation and the material itself.

4.6 EDA Findings and Preprocessing Implications

During the preliminary exploration, several observations were made which would have an impact on the pipeline design:

Observation	Implication for the Pipeline
Multiple audio recording sources	Audio standardization is required
Noise, reverberation and overlapping speech may occur	VAD and speaker-aware processing are required
Different meeting durations	Meetings need to be processed independently
Multiple dialogue-act categories	Transcript analysis should consider conversational context
Decision annotations are available	They can provide reference information for decision analysis
Presentations may use legacy `.ppt` format	Format conversion is required before extraction
Audio and document data contain complementary information	Cross-modal alignment is required

Overall, the EDA was performed as a pre-processing and modality-specific data exploration process rather than regular tabular EDA.

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